

SIERRA AI Literacy Program

22-Day Learning Journey (April 1 - May 1, 2026)

WEEK 1 (April 1-3): First Steps

3 days: Wednesday - Friday

Day 1: Why Sierra

- Sierra is the ABS-approved AI assistant — uses best-available models, resides on ABS IT infrastructure
- Protects client data and keeps messages private from AI vendors
- Meets compliance requirements
- Other AI assistants: not explicitly approved, risk of data leakage

Day 2: Privacy, Security & Data Governance

- Policy Compliance: users must abide by the Sierra Data Policy; client contractual terms supersede it
- App Design & Data Governance: chat threads are private; workspaces manage collaboration; limited web search/external resource usage
- Protected IT Infrastructure: approved by ABS cyber teams
- *(Three-layer pyramid: Policy Compliance → App Design & Data Governance → Protected IT Infrastructure)*

Day 3: What Is a Large Language Model?

- An LLM is a calculator that converts a text input (prompt) to a text output (response)
 - It has learned patterns from vast amounts of text to produce replies that seem knowledgeable and logical
 - *(Visual: Text Input → Large Language Model → Text Output)*
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WEEK 2 (April 7-11): AI Fundamentals

5 days: Monday - Friday

Day 4: What the Model Knows — and Where It Stops

- Internal knowledge does NOT include: anything never on the public internet, events after training cutoff, anything ABS-specific
- Internal knowledge DOES include: public knowledge, historical events, general logic, deep domain expertise
- Training cutoffs for current Sierra models: late 2024 to mid-2025
- Rule of thumb: if it was never on the public internet, it is not in the model's internal knowledge
- Everything ABS-specific must come from the user through context

Day 5: Context — The Lever You Control

- Two and only two sources: internal knowledge (fixed) and context (what you provide)
- Context is the lever users control — internal knowledge is not

- Context includes: prompt text, files you attach, conversation history
- The model works with both simultaneously — context shapes what it reasons over
- Goal: load up context with what the model can't know on its own

Day 6: SIERRA's Tools for Adding Context

- Sierra is not a model — it's a platform with secure access to various LLMs and a set of tools built around them
- The models were trained on public data; ABS-specific work — projects, rules, client data, procedures — is not in there by default
- Sierra has three tools to bring in what the model doesn't know: web search (current/external info), file attachments (documents you upload), and knowledge repositories (CoE-built data sets Sierra searches automatically)
- Example: HR vacation policy — the model has no idea; Sierra retrieves it from an ABS-built knowledge repository and responds correctly
- In all three cases, the model only processes what's in its context window — these tools exist to make sure the right information gets there

Day 7: Tokens & Context Windows

- The model only processes what's in its context window — nothing from other threads, nothing not explicitly provided or retrieved
- Context window = working memory: everything the model can hold and process at once — large, but not infinite
- Tokens are the unit of measurement: roughly 666 tokens per page
- The quality constraint: the model processes everything in the window on every message — irrelevant content degrades output quality, not just efficiency
- Focused threads get better results: start a new chat when switching topics, keep threads tight and on one subject
- Failure mode: long conversations push earlier context out of the window — the model loses information it once had; that sets up Day 8

Day 8: What Is a Hallucination?

- A hallucination is a confident, fluent output not grounded in truth
- Models are trained to produce the most probable continuation of text, not necessarily the most truthful
- This often occurs with insufficient context — filling a gap based on patterns, not real data
- This is why much of the focus is on building internal knowledge systems
- Also watch for: out-of-date responses (verify anything time-sensitive against the cutoff) and irrelevant responses (if Sierra misses the point, add more context)
- Always review outputs before sending to clients — fluent does not mean accurate

WEEK 3 (April 14-18): Prompting & How Sierra Works

5 days: Monday - Friday

Day 9: Prompting Fundamentals

- Vague in = vague out — specificity is the single biggest lever

- Four ingredients: Role (the angle of approach — "you are a technical reviewer"), Task (explicit instruction — the more specific and example-backed, the better), Context (background information and raw material the model couldn't know on its own), Format (output specification: bullets, table, one page, plain language)
- Iterative by design — unlike a search engine, which ends at the first result, the conversation continues; build on what it returns until you get what you want
- Your first prompt is a starting point — when Sierra misses, tell it what was wrong and add what it was missing; when a thread goes sideways entirely, start fresh
- Meta-prompt shortcut: ask Sierra to write the prompt for you — describe the task rough, stream of consciousness, and ask it to structure a clean prompt from what you provided

Day 10: Structuring Complex Tasks

- You are the SME — Sierra does not know your process, project, client, or professional judgment; your job is to direct it, not hope it figures it out
- Don't throw a complex problem at it all at once — break it into sequential steps and guide it through
- One big prompt = one shot; sequential steps = you review each output before the next starts, catching errors early
- You decide the sequence and what good output looks like at each stage
- Once a workflow is dialed in, ask Sierra to build a reusable prompt from the conversation history — save it as a starting point for next time

Day 11: Structuring Prompts for Data Extraction

- What you are leveraging here is processing power, not reasoning — the model's ability to read and ingest massive amounts of information and pull out key data at scale; the goal is deterministic output, not brainstorming
- The organizing principle: reduce the model's cognitive load — every ambiguity you leave unresolved is left to the model to fill; the less it has to figure out, the more it focuses on the extraction
- Pre-prompt checklist: Field (what am I extracting?), Definition (what exactly counts?), Exclusions (what does not count?), Location (where in the document is it found?), Decision rules (if/then logic for ambiguous cases), Evidence (what proof should Sierra return?), Output format (how should results be structured?)
- Encode the structure — map your checklist to the document before you write the prompt; name the section, heading, or clause where each field lives
- Phased extraction: Pass 1 — pull high-level anchors (party names, vessel type, applicable schedule); Pass 2 — inject those anchors back into the prompt with the same file and precise instruction for granular fields; the second pass is grounded, not guessing
- ABS example: Pass 1 — "Identify the contracting party, contract type, and which pricing schedule applies." Pass 2 — "Under Schedule B as confirmed above, extract the day-rate for each vessel class as defined in Clause 4. List exclusions. Return one row per vessel class with: vessel class | day-rate | applicable clause | exclusions."

Day 12: Working with Files — Part 1

- Common misconceptions: uploaded files are not added to Sierra's internal knowledge; the entire file is not dumped into the context window on every message

- What actually happens: file is converted to raw text and indexed into a separate store; when you send a query, Sierra pulls the most relevant sections into context and sends those — along with your question — to the model; different queries surface different sections
- Preview is available before attaching — you can inspect what the conversion produced before it hits the thread
- Source quality matters — clean, structured text converts and retrieves more reliably; complex, image-heavy files produce less reliable conversions
- Thread attachments are temporary and isolated; workspace files are persistent and shared across all users

Day 13: Working with Files — Part 2

- No amount of prompting compensates for data that wasn't read in correctly — input integrity comes before prompt quality
 - What to consider before attaching: Is it text-based or image-heavy? Is it a scanned PDF or a DOCX? Tabular or diagram-heavy? These determine conversion quality
 - How conversion works: images and diagrams are converted to text descriptions; complexity affects how accurate and complete that description will be
 - Fast vs. full conversion: by default Sierra uses fast (strips to raw text, maintains general structure); switch to full for scanned or image-heavy files — full runs OCR first and produces a more complete, structured output
 - Preview the conversion — check it before relying on it; if the conversion is wrong, the prompt won't fix it
 - The default troubleshooting instinct should be the same as a calculator: if the output is wrong, check what was keyed in — the problem is usually in the data ingested, not the model
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WEEK 4 (April 21-25): Knowledge Systems & Tools

5 days: Monday - Friday

Day 14: Memories

- Memories are persistent context — injected into every thread automatically, no retyping required
- What to put in them: role and specialization; operational context (industry, region, team, organization); preferences (writing style, output format)
- Set them by telling Sierra directly ("Remember that I...") or through profile settings
- Save key context once — automatically included in every thread; the model output is grounded and more tailored before you type a word

Day 15: Workspaces

- A workspace is a collaborative environment where multiple members can access common instructions, context, and files
- Workspace Context: background knowledge that applies across all threads in the workspace
- Workspace Memories: shared information relevant to the workspace — can be dynamically added by each member
- Workspace Files: searchable/retrievable knowledge sources available across threads and members — mention file names in prompts to improve retrieval

- Workspace context is shared and persistent; thread content is yours and temporary

Day 16: Workspace Task Modes

- Task Modes: reusable step-by-step instructions for specific, repeatable workflows — invoked by users when needed
- What a Task Mode can include: steps, which workspace files to reference, which tools to activate, output format requirements
- Rule of thumb: general background → Workspace Context / specific repeatable activity → Task Mode
- Real examples: emergency response validation, incident root cause analysis, regulatory compliance check, flag state compliance
- Context + Memories are always on for everything; Task Modes are invoked when needed for specific jobs

Day 17: Workspace Walkthrough

- Live walkthrough of the Sierra workspace interface: Settings (name, description, member access), Context (workspace-level background available in every thread), Files (uploaded once, accessible across all chats and members), Task Modes (saved reusable prompts for specific tasks), and Memories (in beta)
- Personal memories: your role, specialization, and preferences — injected across all your personal chats
- Workspace memories: project/client-specific context — shared with all workspace members, not the whole company
- Setting workspace context means every new thread within that workspace starts already up to speed on the domain

Day 18: Workspace Development

- Start building in a regular chat, not the workspace — describe what you're configuring at a high level (can use mic/stream of consciousness), then let Sierra help draft the components
- Step 1: establish shared understanding — describe the workflow, attach notes, ask Sierra to summarize to verify it understands the domain before building
- Step 2: analyze source material — upload samples and templates, ask Sierra to extract structure, variables, boilerplate vs. customization, and patterns; review before proceeding
- Step 3: generate workspace context — ask Sierra to write the context block including purpose, structure, variables, and an onboarding section for new members
- Step 4: build the task mode — describe the steps you want the prompt to take, let Sierra write the instructions, paste into the workspace
- Step 5: upload workspace files — templates and reference documents loaded once, available to all chats
- The 80/20 principle: Sierra handles 80% of the work; teams finish the last 20%

WEEK 5 (April 28 - May 1): Live Demos

4 days: Monday - Thursday

Day 19: Plato Hull Survey Demo

- Live demo of Plato answering a multi-part rule interpretation question: annual hull survey scope, suspect areas, and close-up survey expectations for a bulk carrier

- Plato returns structured answer with citations and clickable links to ABS Survey After Construction Rules (Part 7, Chapter 3, Section 2), distinguishing between non-double-skin and double-skin bulk carrier requirements
- Follow-up scenario: suspect area with substantial corrosion and coating breakdown in a ballast tank — Plato connects two rules and returns a survey prep game plan (re-examination, thickness measurements, potential gauging extension)
- References go to subsection level (1.1.9 suspect areas, 1.1.10 ballast tank requirements)
- Compliance note: usage must follow the Sierra by Engineers and Surveyors guide issued by the Chief Surveyor and Chief Engineer

Day 20: Environmental Plan Review Workspace

- Live build of a workspace for engineering environmental plan reviews — context generated by Sierra, pasted into workspace; each plan type gets its own task mode
- Workspace context: general environmental plan review guidance, file-only sourcing (no internet), output style — generated in Markdown for easy LLM consumption
- Task mode built for SOPEP review: 34 existing review questions fed into Sierra, which generates the full task mode instructions; pasted into workspace
- Demo run: SOPEP file uploaded, task mode invoked, output includes summary and a compliance checklist (yes/no per question, finding, source reference) plus a reviewer check for human sign-off
- Key point: one workspace, multiple task modes — general context and style consistent across all plan types, specific instructions per plan

Day 21: Web Search and Deep Research

- Live demo of web search and deep research tools applied to a current maritime topic: alternative fuel pathways (ammonia, methanol, LNG)
- Step 1 — web search: structured comparison across five dimensions, executive summary, and comparison table with source citations
- Step 2 — judgment layer: trade-offs, risks, and uncertainties per fuel in five bullets; Sierra frames the issue for decision support, separating near-term commercial readiness from long-term decarbonization potential
- Step 3 — deep research: full technical report with detailed citations, downloadable, reusable for briefings and client-facing materials
- Bottom line from demo: LNG most operationally established, methanol has near-term momentum, ammonia has long-term potential but significant hurdles
- Sierra links to evidence throughout — outputs are a starting point for review, not finished conclusions

Day 22: HR Policy Repository

- Live demo of Sierra's built-in HR policy repository covering ABS Bureau, ABS Group, and Wavesight across dozens of countries
- Sierra automatically applies the policy for the user's entity and home country — no need to specify; every employee gets answers tailored to their own profile
- Demos: PTO query (US Group employee), parental leave for Bureau employee in Norway, travel policy for ABS Group US
- Sierra retrieves only from HR documents — no external sourcing — and cites exactly where each answer came from

- For complex or edge-case questions: if the policy isn't in the repository, Sierra says so clearly and directs the user to their HR business partner rather than guessing
 - Series close: certification through iAchieve; reach out to the AI Center of Excellence for support
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Course Structure

Format: 1 module per day, 22 business days total

- 5-minute instructional video
- 1 hands-on exercise
- Final assessment at completion

Schedule: April 1 - May 1, 2026

- Week 1: 3 days (Wed-Fri)
- Weeks 2-4: 5 days each (Mon-Fri)
- Week 5: 4 days (Mon-Thu)